

Jingsong Chen

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EDUCATION

University of Washington, Seattle

GPA: 3.92

Applied and Computational Math Sciences: Discrete Math and Algorithms, BS

Seattle, WA

September 2022 – March 2026

University of California, San Diego

Master of Science in Computer Science and Engineering

San Diego, CA

September 2026 – present

INTERNSHIP

CIMER. Co

Software Engineer

Nanjing, China

Jul 2025 – Sep 2025

- Built a Retrieval-Augmented Generation system with strict tenant isolation and switchable LLM providers (Local Ollama, Gemini, OpenAI).
- Designed a robust ingestion pipeline: multi-format parsing (PDF/TXT/DOC/DOCX), semantic chunking, vectorization, and Weaviate storage.
- Implemented true multi-tenancy using Weaviate tenants; enforced isolation in batch writes/queries, preventing cross-tenant data leakage; refactored Celery tasks to shared parsing/chunking services; fixed parser-chunker data contract to eliminate vectorization failures.
- Integrated vector clustering to accelerate retrieval and improve scalability for large document collections.
- Added citation support, displaying document chunks referenced by the language model for transparency and traceability.
- Shipped dynamic model configuration: frontend settings persisted to backend and synced to Weaviate's generative config for immediate provider switching.
- Supported national cybersecurity competition software and translated technical documentation and client meetings between English and Chinese.

Nanjing University

Data Engineer

Nanjing, China

Jul 2026 – Sep 2026

Crack Layout Model Training and Generative Rendering - *Model Training and Fine Tuning*

- Developed a two-stage synthetic-data pipeline to generate diverse, controllable training data for UAV-based facade crack detection and analysis: (1) designed and trained a custom facade-conditioned Conditional Flow Matching model that integrates crack-free wall RGB images, facade masks, boundary and distance maps, density classes, and random noise to generate continuous crack soft maps encoding skeleton topology; and (2) used a fine-tuned FLUX.1-Fill-dev LoRA to render realistic cracks at the generated locations.
- Built component-aware preprocessing for high-resolution facade images, producing approximately 100,000 aligned 448 x 448 image, ground-truth, edit-mask, and paste-mask samples; grouped splits by source image to prevent data leakage.
- Fine-tuned and deployed FLUX Fill with BF16, LoRA, gradient checkpointing, and multi-GPU model sharding; tuned guidance and LoRA strength to reduce overly dark or distorted generations.
- Implemented Telea-based crack removal, sequential patch synthesis, Lab-space luminance matching, and safe compositing that preserved every pixel outside the target mask while restoring full-resolution outputs.

Crack Data Synthesis and Donor Retrieval - *Algorithm Design*

- Built a Python data pipeline for facade crack synthesis, pairing images and masks by source ID, extracting donor features, and exporting synthetic images with aligned segmentation masks.
- Designed topology-aware donor retrieval using skeleton, geometry, appearance, and spatial features; extracted 1024/1536-pixel network patches to retain local branching structures at usable scales.
- Implemented ranked-candidate fallback and constrained scaling, rotation, flipping, and translation to fit real crack textures within target facades, with coverage checks and per-image donor reuse limits.
- Optimized processing with persistent CUDA/BF16 layout inference, CPU multiprocessing, and component-local skeletonization; added resumable stages and donor-level transformation metadata for reproducible batch generation.

PROJECTS	
Transformer-Based Jet Classification <i>PyTorch • Transformers • Random Forest • Physics-Informed ML • Attention Mechanisms</i>	Seattle, WA May 2025 – June 2025
- Built a Transformer encoder model to classify Large Hadron Collider jet events using variable-length input sequences of particle features, achieving ~77% validation accuracy. - Applied Transformer + MLP Structure and trained a Random Forest classifier on pooled Transformer embeddings to improve interpretability. - Preprocessed and managed a dataset of 50,000 jet events from ATLAS (detectors at the Large Hadron Collider) simulation; implemented model in PyTorch with dynamic padding and training visualizations.	
AutoML-Agent: LLM-Driven Dynamic Model Trainer <i>AI Agent</i>	Seattle, WA May 2025 – June 2025
- Developed an agent-based machine learning pipeline in Python using PyTorch and Hugging Face Transformers to automate dataset analysis and model selection. - Integrated a local LLM to parse natural language queries and recommend optimal model architectures (e.g., CNN vs. linear model). - Implemented custom convolutional and fully connected neural networks, achieving accurate dynamic training on various benchmark datasets.	
Medical Imaging Signal Compression with PCA and Phase Plane Analysis of Body Fat Dynamics <i>Application of Principal Component Analysis</i>	Seattle, WA Jan 2025 – Mar 2025
- Applied Principal Component Analysis to 64-channel detector data from a silicon photomultiplier for signal compression in medical imaging. - Reduced sensor dimensionality from 64 to 9 while preserving signal fidelity, enabling substantial cost savings on electronic readout. - Visualized eigenmodes and interpreted the physical meaning of principal components. - Modeled human body fat regulation using nonlinear ODE systems, where current body-fat levels influence caloric intake and energy expenditure in a feedback loop. - Performed stability analysis using Jacobians and phase portraits, tuning metabolic parameters to find spiral and node behaviors.	
Cancer Care Interactive Information System <i>Application of Data Visualization</i>	Seattle, WA Jan 2025 – Mar 2025
- Developed an interactive information platform with dynamic visualizations, allowing patients, families, and medical providers to explore cancer treatment options, survival rates, and regional healthcare disparities through accessible language and engagingly structured, visual presentations of vital information. - Enabled real-time data filtering and comparison to support evidence-based decision-making and improve understanding of complex healthcare information. - Led comprehensive data preparation using SQL to clean, merge, and enrich large-scale multi-country cancer datasets (incidence, mortality, economic cost etc.), enabling analysis of how healthcare access, environment, and lifestyle risk factors drive disease burden and informing both prevention and policy discussions.	
RELEVANT SKILLS	
- Programming: Python, Java, C++, SQL, OCaml, Scheme, Ruby. - Machine Learning: PyTorch, scikit-learn, Transformers, CNNs, Random Forest, PCA, clustering. - Generative AI: Diffusers, FLUX, LoRA, RAG, Weaviate, Hugging Face Transformers. - Data and Backend: NumPy, FastAPI, Docker, relational and vector databases.	
- Visualization and Tools: Matplotlib, Jupyter Notebook, Observable Vega-Lite, Git. - Data Processing: custom Dataset/DataLoader, dynamic padding, masks, model evaluation.	